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Unsupervised Graph-Based Deep Clustering for Optimal Monitoring Point Placement in Urban Drainage Networks

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ABSTRACT

The optimal placement of monitoring points in urban drainage networks is critical for operational efficiency, yet traditional methods suffer from high costs and redundancy. To address this, we propose a novel unsupervised framework that integrates a graph convolutional network for topological feature extraction with a structured deep clustering network to identify critical monitoring zones without requiring labeled data. On both a self-collected dataset and the public C-Town benchmark, the model demonstrated high performance, achieving over 94% clustering accuracy against expert-defined labels and identifying functional network zones with high precision. This high-precision clustering yields significant operational improvements: the number of required monitoring points was reduced by over 40% while critical node coverage increased to over 90%. This optimized layout also enhanced monitoring reliability by over 8% and shortened pollution event response times by nearly 36%, offering a scalable, data-driven solution for intelligent and efficient urban water management.

1 | Introduction

As urbanization accelerates, the drainage network has become a critical component of urban infrastructure. The stable operation of this network is directly linked to urban flood control, drainage efficiency, water environment quality, and public health [1]. However, pipeline network systems face severe challenges, including large scale, complex topology, and aging facilities, which can lead to blockages and leakages [2]. Consequently, efficient and accurate monitoring has become an urgent task in urban governance. Traditional approaches, such as manual inspection or experience-based fixed-point monitoring, are plagued by high costs, low efficiency, data redundancy, and insufficient coverage, making it difficult to dynamically reflect the network's operational status [3].

In recent years, deep learning techniques, particularly graph neural networks (GNNs), have emerged as powerful tools for analyzing complex graph-structured data, showing great potential for the intelligent analysis of urban pipe networks [4]. Alongside these graph-based approaches, advanced probabilistic and surrogate-based methods are being developed for the reliability analysis of critical infrastructure such as pipelines [5, 6]. Notably, recent work has also focused on Kriging-assisted reliability-based design optimization (RBDO) for large-scale structures like offshore wind turbines, a methodological domain that is highly relevant to the surrogate-based optimization approach proposed in this study [7]. Various studies have successfully applied graph convolutional networks (GCNs), a prominent type of GNN, to different aspects of water and drainage systems [8–18]. In the field of pattern recognition, GCNs and related models

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like GraphSAGE have been used to identify distinct drainage patterns from topological and geometric features [9, 18]. For state prediction and structural health monitoring, a wide range of machine learning and deep learning frameworks are being applied. These include models for predicting fatigue life from material defects, assessing the seismic performance of buildings, assisting in the design of structural components like shear walls, and automating crack detection using computer vision [12–15]. In parallel, for applications within water networks, specialized models such as deep GCNs and Graph-Wavenet have been developed to predict pipe failures and forecast pressure at key sites [10, 11]. Furthermore, GCNs have been integrated into sophisticated frameworks for anomaly detection, such as identifying leakage zones using spatio-temporal GCNs (ST-GCN), and for complex decision-making, such as optimizing post-disaster repair sequences using deep reinforcement learning [16, 17].

However, these approaches suffer from two critical limitations that hinder their practical deployment. First, their supervised learning paradigm requires extensive, high-quality labeled data (e.g., historical failure records), which is often unavailable. Second, they focus on predictive tasks, overlooking the more fundamental engineering challenge of optimal sensor placement [19]. This leaves a critical gap for a structure-aware, unsupervised method that can determine where to monitor to maximize network observability without requiring pre-existing labels. To fill this gap, this study proposes an end-to-end unsupervised framework that leverages a GCN with a structured deep clustering (SDC) network to achieve an efficient and scientific layout of monitoring points.

The main contributions of this study are as follows:

1. *A novel end-to-end unsupervised framework:* To the best of our knowledge, this study introduces the first end-to-end unsupervised framework that synergistically combines GCNs with SDC for monitoring point optimization, critically eliminating the dependency on costly and often unavailable labeled data.

2. *Novel dual self-supervision mechanism:* We propose an innovative dual self-supervision mechanism that learns highly discriminative network representations by jointly optimizing clustering assignments and feature reconstruction, leading to more robust and accurate cluster identification.
3. *Comprehensive empirical validation:* We provide extensive empirical evidence of our method's superiority, demonstrating through rigorous experiments on both real-world and benchmark datasets that it significantly outperforms traditional and state-of-the-art (SOTA) deep graph clustering approaches in accuracy, efficiency, and optimization effectiveness.

2 | Methods and Materials

This section first introduces the GCN-based topological feature extraction method for pipeline network nodes. Then, it elaborates on the monitoring point optimization strategy that incorporates a SDC network, and finally describes the dataset and environment configuration used for the experiment.

2.1 | GCN-Based Topology Feature Extraction for Drainage Network Node

A drainage network is essentially a complex graph structure, and the topological relationships between nodes are critical to the overall functionality of the system. GCN can directly process irregular graph data (Figure 1) and learn the features of a central node, which are organized into a node feature matrix (FM) by aggregating neighboring node information. This makes it ideal for extracting features from pipeline networks [20]. GCN can handle both regular data structures where the nodes are arranged in a regular grid (Figure 1a) and non-regular data structures closer to the actual pipe network where the nodes are arranged irregularly and have complex connections (Figure 1b).

The fundamental principle of GCN is to update a node's feature by combining data from its neighbors [21]. Our study adopts a

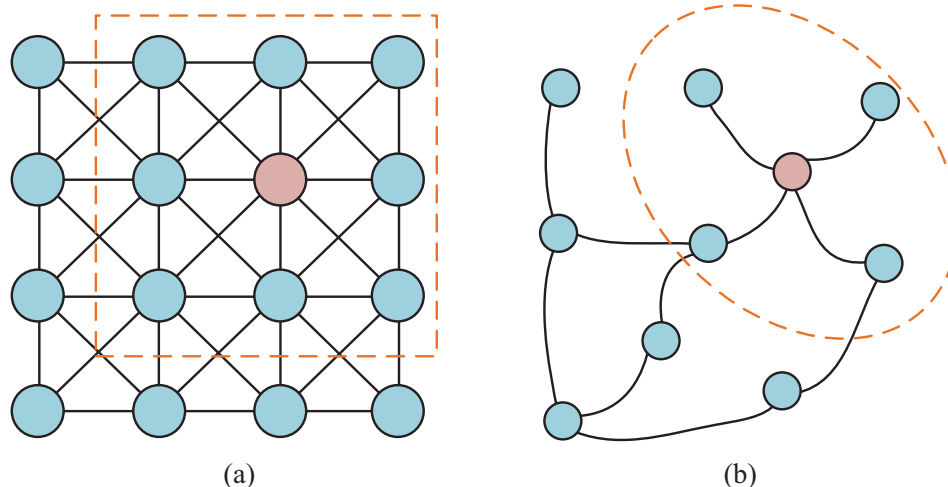


FIGURE 1 | Structure diagram of (a) regular data and (b) irregular data structures in GCN.

multi-layer GCN structure, with the layer-wise propagation rule shown in Equation (1).

$$\begin{cases} H^{(g+1)} = \sigma(\hat{A}H^{(g)}W^{(g)}) \\ \hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}} \end{cases} \quad (1)$$

Conceptually, this rule states that a node's features are updated by aggregating and transforming the features of its neighbors from the previous layer. In Equation (1), A is the adjacency matrix and I is the identity matrix. The term $\tilde{A} = A + I$ represents the adjacency matrix with self-loops added to each node. $\hat{A}$ denotes the normalized adjacency matrix. $H^{(g+1)}$ denotes the node FM of the $g + 1$ th layer. $\tilde{D}$ denotes the degree matrix of $\tilde{A}$. $H^{(g)}$ and $W^{(g)}$ denote the feature node matrix and weight matrix (WM) of the g th layer, respectively. σ denotes the nonlinear activation function.

The GCN's structure is shown schematically in Figure 2. The input layer (IL) represents the initial node features. The hidden layer (HL) applies graph convolution operations to generate new feature representations. Finally, the output layer (OL) produces the final result. In constructing the basic computational unit of GCN, the propagation and aggregation of node information through normalized adjacency matrix and feature transformation are investigated [22].

To prevent the "over-smoothing" issue in deep GCNs and retain node-specific characteristics, we introduce a feature representation from a self-encoder, which is combined with GCN features via linear interpolation, as shown in Equation (2) [23].

$$\begin{cases} F^{(g)} = \phi(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}F^{(g-1)}W^{(g-1)}) \\ \tilde{F}^{(g-1)} = (1 - \epsilon)F^{(g-1)} + \epsilon H^{(g-1)} \end{cases} \quad (2)$$

In Equation (2), ϕ denotes the activation function. $\tilde{F}^{(g-1)}$ denotes the FM that combines the output of the self-encoder and the output of the GCN layer. ϵ denotes the equilibrium coefficient.

The learned features are then converted into probability distributions using the Softmax function for subsequent clustering, as shown in Equation (3) [24].

$$F' = \text{softmax}(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}F^{(G)}W^{(G)}) \quad (3)$$

In Equation (3), F' denotes the final output FM. $F^{(G)}$ and $W^{(G)}$ denote the output FM and WM of the final layer, respectively.

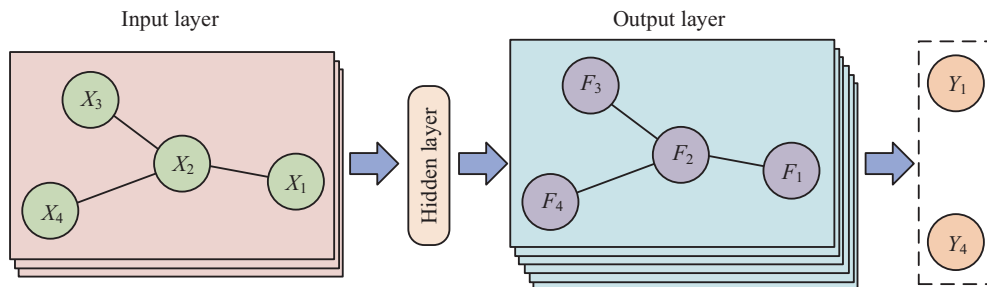


FIGURE 2 | Schematic diagram of graph convolution structure of GCN.

Figure 3 illustrates this overall feature extraction process.

In Figure 3, drainage network node topological features are extracted and transformed step by step through multilayer processing of GCN. By combining the data from nearby nodes via a graph convolution process, the node's feature is updated at each stage. After multilayer processing, a high dimensional feature of node topology and attribute features is obtained.

2.2 | The SDC Framework

To identify key node clusters from the high-dimensional features, we propose a novel framework named the SDC network. The SDC model is an end-to-end unsupervised method composed of three key modules: a deep neural network (DNN) for non-linear feature mapping, a GCN for capturing topology, and a dual self-supervised module for joint optimization of clustering assignments and feature reconstruction. The overall architecture is detailed below.

2.2.1 | Model Architecture

The DNN module serves as a non-linear encoder to map the initial high-dimensional features into a more compact and cluster-friendly latent space. It is structured as a multi-layer perceptron (MLP) with the following architecture: $d \rightarrow 256 \rightarrow 64 \rightarrow K$, where d is the dimension of the input features from the GCN, and K is the number of target clusters. We use the rectified linear unit (ReLU) activation function for all HLs. The final output Z (as shown in Equation (4)) represents the low-dimensional embedding of each node [25, 26].

$$Z^{(l)} = \sigma(W_{\text{DNN}}^{(l)}Z^{(l-1)} + b_{\text{DNN}}^{(l)}) \quad (4)$$

In Equation (4), $Z^{(l)}$ denotes the low-dimensional FM output from the l th layer. $W_{\text{DNN}}^{(l)}$ and $b_{\text{DNN}}^{(l)}$ denote the WM and bias vector of the l th layer.

2.2.2 | Joint Loss Function

The dual self-supervised module optimizes the model through a joint loss function L that combines a clustering loss L_{k_l} and a feature reconstruction loss L_{rec} . This ensures that the learned embeddings are not only well-separated but also retain essential

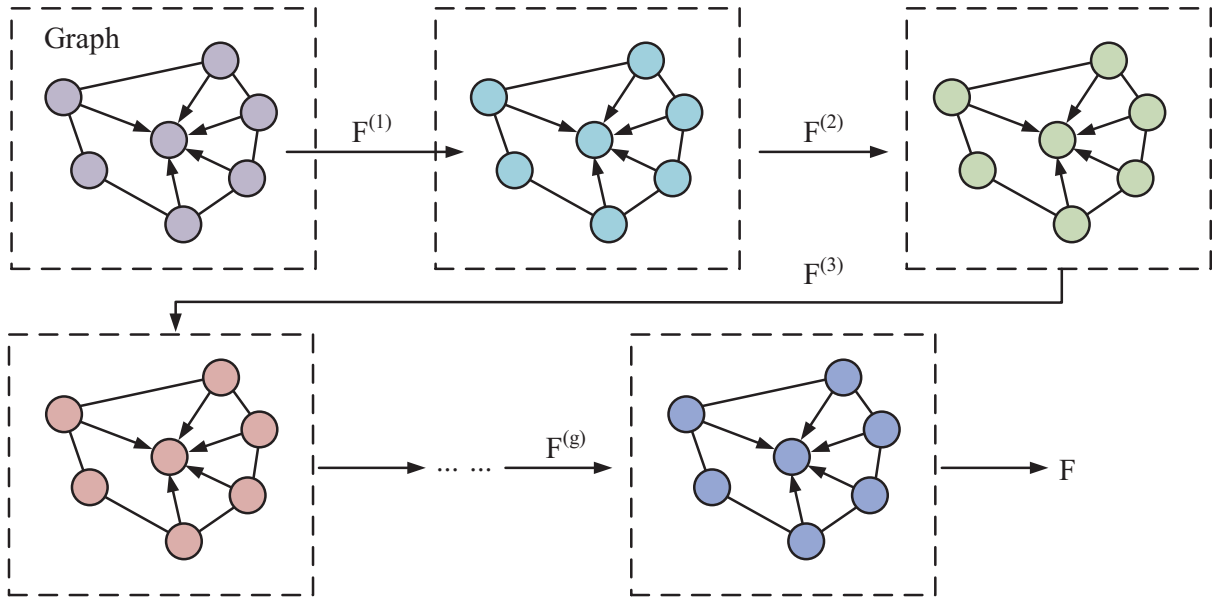


FIGURE 3 | Schematic diagram of the GCN-based topological feature extraction process for drainage network nodes.

information from the original features. The total loss is defined as Equation (5).

$$L = L_{k_i} + \lambda \times L_{rec} \quad (5)$$

where λ is a hyperparameter that balances the two objectives. In our experiments, the hyperparameter λ was optimized via a grid search over the values $\{0.01, 0.1, 1.0, 10.0\}$. Using a 20% validation split of the self-collected dataset, we found that $\lambda = 0.1$ provided the optimal balance between clustering performance (as measured by the silhouette coefficient) and feature information preservation (L_{rec}), and this value was used for all experiments.

The clustering loss L_{k_i} is calculated using the Kullback–Leibler (KL) divergence to encourage the learned node embeddings (h_i) to converge toward defined cluster centers (μ_j).

First, we compute the soft assignment distribution Q , which represents the probability of assigning node i to cluster j using the Student's t -distribution, as proposed in Equation (6) [27]. This distribution is particularly effective as it is robust to outliers and its heavier tails help to better separate clusters by assigning higher confidence to nodes closer to the cluster centers.

$$q_{ij} = \frac{\left(1 + \|h_i - \mu_j\|^2\right)^{-1}}{\sum_{j'=1}^k \left(1 + \|h_i - \mu_{j'}\|^2\right)^{-1}} \quad (6)$$

In Equation (6), q_{ij} denotes the soft assignment probability of node i to cluster j . μ_j denotes the center of cluster j . h_i denotes the feature vector of node i . k denotes the total number of clusters.

To refine the clusters and accelerate convergence, we then define an auxiliary target distribution P by sharpening Q , which helps the model learn from high-confidence assignments, as shown in Equation (7) [28].

$$p_{ij} = \frac{q_{ij}^2 / \sum_i q_{ij}}{\sum_{j'} q_{ij'}^2 / \sum_i q_{ij'}} \quad (7)$$

In Equation (7), P_{ij} denotes the probability that node i in the target distribution belongs to cluster j .

The clustering loss L_{k_i} is the KL divergence between the auxiliary distribution P and the soft assignment distribution Q , as shown in Equation (8).

$$L_{k_i} = KL(P \| Q) = \sum_i \sum_j p_{ij} \times \log \left(\frac{p_{ij}}{q_{ij}} \right) \quad (8)$$

The reconstruction loss L_{rec} ensures that the low-dimensional embeddings retain the core information of the original high-dimensional features F . A decoder network is used to reconstruct the features $\hat{F}$ from the final latent embeddings $Z^{(L)}$, as proposed in Equation (9) [29].

$$\hat{F} = \sigma(W_{dec} Z^{(L)} + b_{dec}) \quad (9)$$

In Equation (9), $\hat{F}$ denotes the reconstruction FM, While W_{dec} and b_{dec} denote the WM and bias vector of the decoder, respectively.

The reconstruction loss L_{rec} is then the mean squared error (MSE) between the original features F and the reconstructed features $\hat{F}$, as proposed in Equation (10).

$$L_{rec} = \frac{1}{N} \sum_{i=1}^N \|F_i - \hat{F}_i\|^2 \quad (10)$$

By minimizing this joint loss function (Equation 5), the SDC network simultaneously learns discriminative cluster assignments and feature representations in an end-to-end, self-supervised manner. In response to the issue of MSE amplifying outlier effects when node features exhibit heteroscedastic noise, several robust alternative loss functions were considered in the study, which can be integrated into the proposed framework while minimizing architecture changes: (1) L1 Loss: The L1 loss is less sensitive

to outliers compared to MSE, as it penalizes errors linearly rather than quadratically. This characteristic makes it more robust in the presence of heteroscedastic noise. (2) Huber Loss: This loss function provides a compromise between MSE and MAE. It behaves quadratically for small errors and linearly for large errors. (3) Correlation-Based Losses: For tasks where preserving the relative relationship or ranking between feature dimensions is more critical than exact value reconstruction, maximizing the correlation between the original and reconstructed features could be a more suitable and robust objective.

Figure 4 displays the overall model's structure. By jointly optimizing the clustering loss and reconstruction loss, the model is driven to learn better clustering results.

In Figure 4, the DNN module undergoes a multilayer neural network process to extract nonlinear features. The GCN module undergoes multi-layer graph convolution operation to enhance the topological features of the nodes. The dual self-supervision module optimizes the clustering effect of the model through dual supervision of clustering assignment and feature reconstruction.

To assess the clustering effect, this study uses silhouette coefficient and Davies-Bouldin index as evaluation indexes [30, 31]. Equation (11) illustrates how the silhouette coefficient is

calculated.

$$\begin{cases} s(i) = (b(i) - a(i)) / (\max\{a(i), b(i)\}) \\ S = \frac{1}{N} \sum_{i=1}^N s(i) \end{cases} \quad (11)$$

In Equation (11), $s(i)$ denotes the silhouette coefficient of node i . $a(i)$ denotes the average distance from node i to other nodes in the same cluster, reflecting intra-cluster tightness. $b(i)$ denotes the minimum average distance from node i to all nodes in other clusters, reflecting inter-cluster separation. S denotes the silhouette coefficient.

After obtaining the optimized clustering partitions, the dynamic correlation between the nodes within each cluster is calculated using the mutual correlation function (Equation 12). The node with the highest average correlation is selected as the monitoring point for that zone. This comprehensive workflow, depicted in Figure 5, ensures that the final layout of monitoring points is optimized based on both the structural properties and dynamic correlations within the network [32].

$$CCF_{ij}(\tau) = \frac{1}{T} \sum_{t=1}^{T-\tau} \tilde{x}_i(t) \cdot \tilde{x}_j(t + \tau), \quad \tau \in [-M, M] \quad (12)$$

In Equation (12), $CCF_{ij}(\tau)$ denotes the mutual correlation function of node i and node j under time lag τ . τ denotes the time

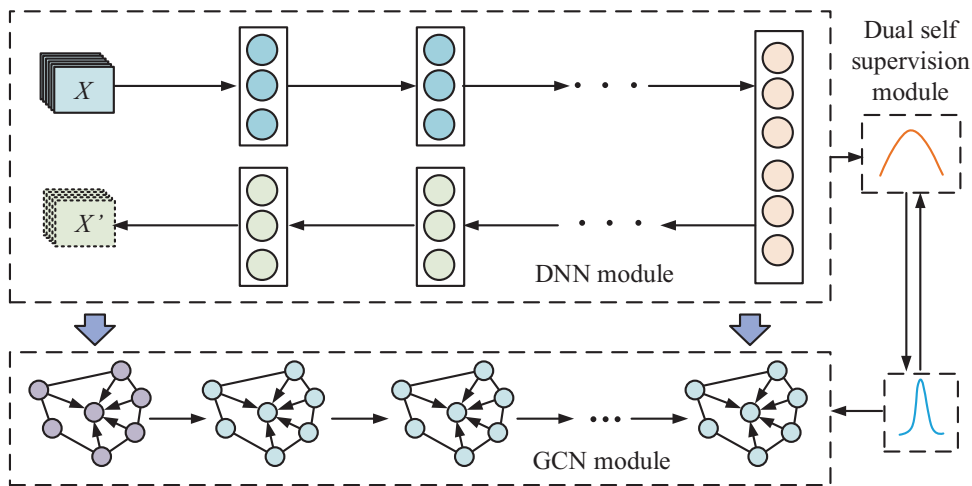


FIGURE 4 | Schematic diagram of the structure of the deep clustering network model.

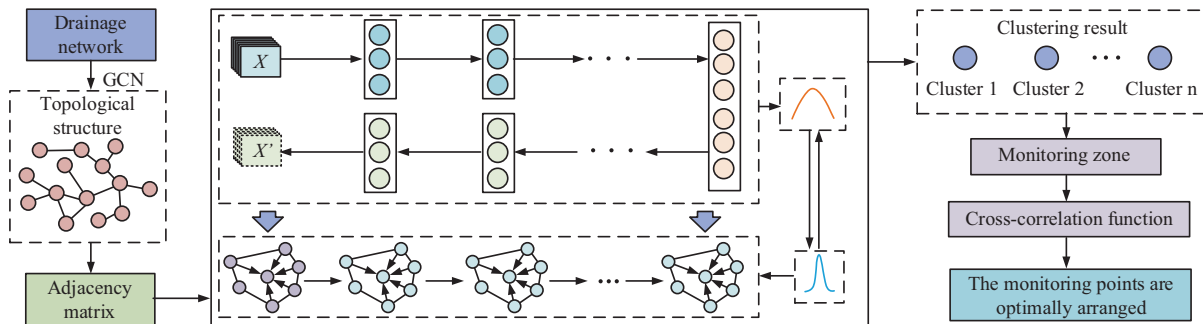


FIGURE 5 | The overall workflow for optimizing the layout of monitoring points using the proposed SDC network model.

lag. M denotes the maximum time lag to capture possible delay effects. $\tilde{x}_i(t)$ denotes the normalized value of node i in time.

In Figure 5, the proposed method first generates the adjacency matrix from the topology of the drainage network, and extracts the high-dimensional feature of the nodes by DNN module and GCN module. Clustering analysis is performed using the dual self-supervision module to obtain the clustering results. Finally, to determine the ideal configuration of monitoring points, the most representative nodes are chosen to serve as monitoring points in each monitoring region.

2.3 | Experimental Setup

This study utilizes two datasets for experimental validation: a self-collected dataset and the public C-Town benchmark.

2.3.1 | Self-Collected Dataset

This dataset was gathered over a continuous six-month period (January 1–June 30, 2025) from a residential drainage network in Lu'an City, China. The network comprises 120 monitoring nodes equipped with ultrasonic level sensors and Doppler flow meters, connected by 156 pipe segments. Data on water level and flow rate were recorded at 5-min intervals. The raw time-series data was preprocessed to handle minor data gaps (approximately 3% of readings were missing due to sensor maintenance or transmission errors), for which linear interpolation was applied. Subsequently, the high-resolution data was aggregated to derive the five time-invariant statistical features for each node, as detailed in Table 1.

To establish a robust ground truth for evaluating clustering performance, the network was manually partitioned into $K = 5$ distinct functional zones. These five zones correspond to distinct hydraulic sub-systems identified by the engineers: two primarily residential areas, one industrial sector, and two mixed-use commercial districts. This partitioning was conducted independently by two senior drainage system engineers, each with over 15 years of field experience, based on a comprehensive assessment of geographic layout, hydraulic connectivity, and documented land use maps. The initial, independent partitions demonstrated high inter-rater reliability (Cohen's Kappa > 0.9). All minor discrepancies, affecting fewer than 5% of nodes, were subsequently resolved in a joint review session to produce a final, definitive ground-truth classification for robust model evaluation. To

mitigate the risk of overfitting on the limited-scale self-collected dataset and enhance model generalizability, we employed a suite of regularization techniques in addition to early stopping at 500 epochs. These included: (1) DropEdge, where 20% of graph edges were randomly dropped during GCN training to improve structural robustness; (2) Feature Masking, where 10% of node features were randomly masked in each iteration to encourage the learning of more robust representations; and (3) Weight Decay (L2 Regularization), applied to all trainable parameters to penalize large weight values.

2.3.2 | Public Dataset

The public dataset is the C-Town network, a widely used benchmark from the “Battle of the Water Calibration Networks” [33]. It represents a real-world, medium-sized network containing 388 nodes and 429 pipes. As it lacks intrinsic node features, we used Node2Vec to generate 64-dimensional feature embeddings for each node. Node2Vec was chosen over other graph embedding methods like DeepWalk due to its superior effectiveness in capturing both local neighborhood (homophily) and broader structural roles (structural equivalence). Its tunable random-walk strategy provides a more flexible approach to defining node neighborhoods, making it uniquely suited for identifying distinct functional zones in complex networks. Unlike purely local methods, its tunable random-walk strategy allows it to interpolate between learning embeddings based on community structure (homophily) and functional roles (structural equivalence), making it uniquely suited for this task. These embeddings served as the initial input for the proposed method. For the purpose of evaluating clustering accuracy (CA), we utilized the ground-truth functional zones established in the original ‘Battle of the Water Calibration Networks’ benchmark. These zones, which correspond to the network’s predefined district metered areas (DMAs), provide a standard basis for performance comparison. The experimental environment and key hyperparameter settings are shown in Table 2. To guarantee the validity of the findings, each comparative experiment is conducted 10 times independently, and the outcomes are averaged.

2.3.3 | Data Processing

2.3.3.1 | Data Preprocessing. For both datasets, the adjacency matrix A was constructed based on the physical pipe connections, where $A_{ij} = 1$ if a pipe exists between node i and j , and 0 otherwise. All node features were normalized using min-max scaling to a range of $[0, 1]$ before being input to the model.

2.3.3.2 | Evaluation Metrics. The metric “key node coverage” was calculated to assess the optimization’s effectiveness in monitoring critical infrastructure. For this study, key nodes were defined as the top 15% of nodes with the highest betweenness centrality, representing the most critical junctions for network-wide flow and distribution.

To assess the clustering effect, this study uses the silhouette coefficient (higher is better) and the Davies-Bouldin index (lower is better) as evaluation indexes. In addition, to quantify the quality of the learned features, we assessed the quality of the learned

TABLE 1 | Features for nodes in the self-collected dataset.

Feature name	Description
Mean flow rate	Average water flow m^3/s over the observation period.
Flow rate Std. Dev.	Standard deviation of water flow.
Mean water level	Average water level (m).
Max water level	Maximum recorded water level (m).
Node centrality	Calculated using the betweenness centrality algorithm.

TABLE 2 | Experimental environment and parameter settings.

Experimental environments		Parameter settings	
Projects	Configuration	Parameters	Values
Operating system	Ubuntu 20.04 LTS	Learning rate	0.001
Processor	Intel Xeon Gold 6248R (3.0 GHz)	Batch size	64
Memory	128 GB DDR4	Iterations	500
GPU	NVIDIA RTX 3090	GCN layers	3
Storage	1 TB NVMe SSD	HL dimension	64
Deep learning framework	PyTorch 1.9.0	Number of clusters	5

representations using Embedding Quality via Linear Probing (a standard protocol where a simple linear classifier is trained on the frozen, learned node embeddings to evaluate their feature separability). For this, a simple linear classifier was trained on the frozen node embeddings generated by the proposed method to predict the ground-truth functional zones. High classification accuracy in this downstream task serves as a robust proxy for the discriminative power and separability of the learned features.

3 | Result

This section presents the experimental results for model performance validation and monitoring point optimization.

3.1 | Model Performance Validation

The study first evaluates the quality of the topological features extracted by the proposed method. This is accomplished by training a linear classifier on the learned node embeddings to predict ground-truth functional zones, where high classification accuracy serves as a proxy for feature discriminability. It is also compared with the current mainstream methods including deep belief network (DBN), convolutional neural network (CNN) and recurrent neural network (RNN). To ensure a fair comparison, the graph structure was first converted into a feature-based representation for these non-graph-native models. We generated 128-dimensional node embeddings using the Node2Vec algorithm, which were then used as the input sequences for the DBN, CNN, and RNN. This approach ensures a fair comparison by encoding the same rich topological information available to our graph-native model into a format suitable for these non-graph-native baselines. In Figure 6, in the self-collected dataset, when the number of iterations is 200, the classification accuracy achieved using the extracted features from DBN, CNN, RNN, and the proposed method (labeled as “SDC” in figures) are 86.37%, 89.82%, 93.35%, and 96.81%, respectively. The proposed method consistently achieved the highest accuracy and demonstrated the fastest convergence rate among all compared models. In the C-Town dataset, when the number of iterations is 200, the feature extraction accuracies of the four methods are 84.87%, 87.65%, 92.15%, and 94.32%, respectively. These findings demonstrate that the proposed method provides superior performance in extracting drainage network topological features,

yielding both higher accuracy and more rapid convergence than baseline models.

To verify the classification accuracy of the proposed SDC network model for drainage network node, the study analyzes its CA in two datasets. It is also compared with existing clustering methods including *K*-means clustering (KMC), spectral clustering (SC), deep embedded clustering (DEC), agglomerative nesting (AGNES). The proposed method (SDC) achieves a stable accuracy of 94.77% on the self-collected dataset, outperforming traditional methods like *K*-means by over 10 percentage points and converging faster than other deep clustering techniques (Figure 7a). This consistent outperformance across methods strongly suggests that the proposed method’s explicit use of the network’s graph topology is a critical factor in achieving accurate node categorization. This consistent outperformance was also observed on the C-Town dataset (Figure 7b), confirming the superiority of the proposed method for drainage network node categorization.

The quality of the resulting clusters was quantified using the silhouette coefficient and Davies-Bouldin index (Figure 8). On the self-collected dataset, our proposed SDC method demonstrated superior cluster quality, achieving a silhouette coefficient of 0.791. This indicates significantly tighter and more well-separated clusters compared to all other methods, including traditional *K*-means (0.428) and the next-best deep clustering method, AGNES (0.705). The Davies-Bouldin index is 0.796, 0.712, 0.629, and 0.575, respectively, showing a clear trend of improving cluster quality with more sophisticated methods. In contrast, our proposed method achieved a superior silhouette coefficient of 0.791 and a lower Davies-Bouldin index of 0.426, indicating the best intra-cluster tightness and inter-cluster separation among all tested methods. A consistent trend is also observed on the C-Town dataset. The silhouette coefficients for the five clustering methods are 0.392, 0.432, 0.527, 0.625, and 0.747, respectively. The Davies-Bouldin index is 0.831, 0.724, 0.629, 0.551, and 0.487, respectively.

To investigate the sensitivity of the results to *K* when the city size expands or the topology structure becomes fractal, an additional experiment was conducted to simulate larger network sizes and increased topological complexity. The results are shown in Table 3. In Table 3, in the scenario of scale expansion, the performance is optimal when *K* = 6, with a CA of 93.05% and a silhouette coefficient of 0.778. In the fractal topology scenario, the

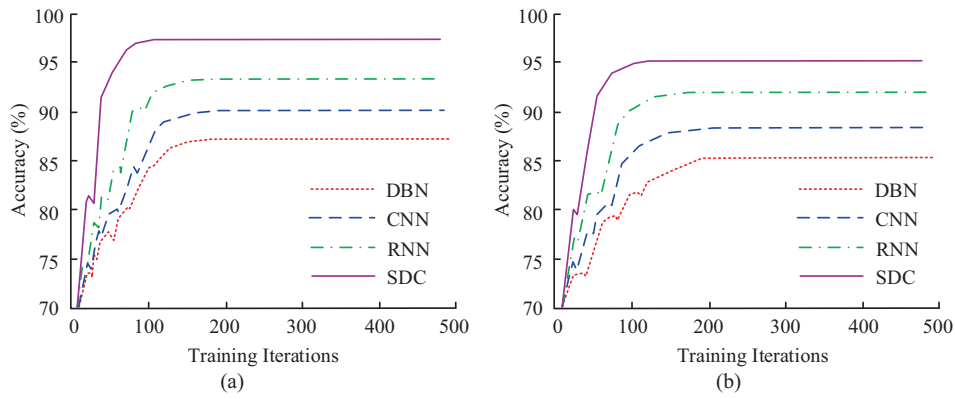


FIGURE 6 | Comparison of topological feature extraction accuracy on (a) the self-collected dataset and (b) the C-Town dataset.

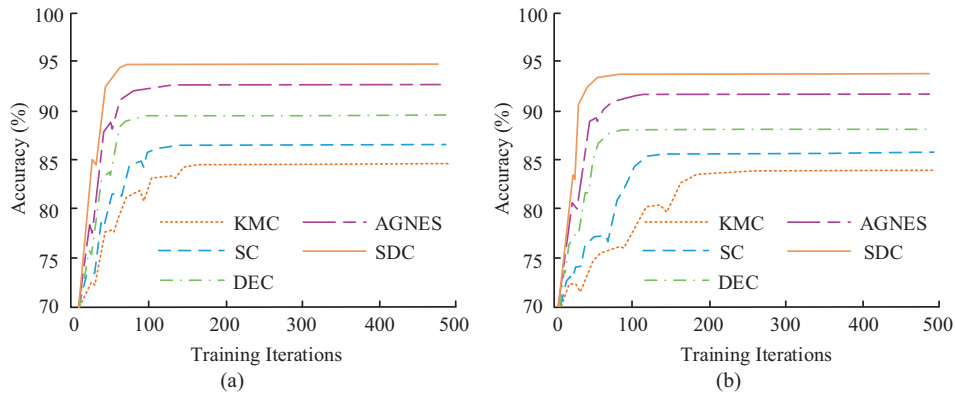


FIGURE 7 | Comparison of CA of different methods on (a) the self-collected dataset and (b) the C-Town dataset.

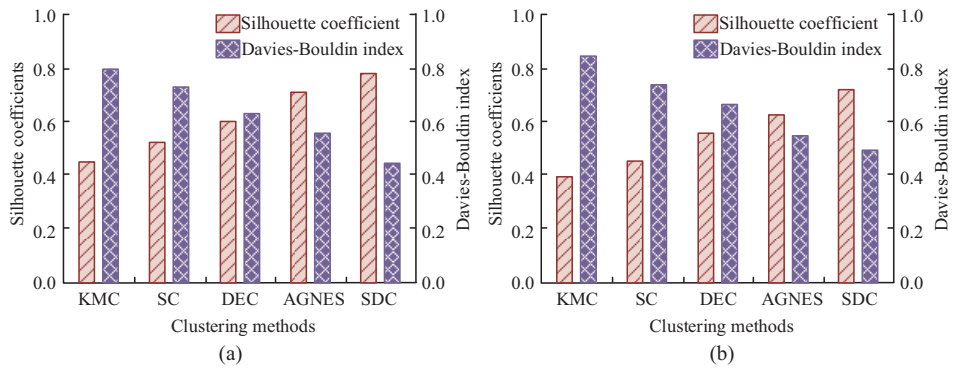


FIGURE 8 | Performance comparison of clustering methods using silhouette coefficient and the Davies-Bouldin index. Results are shown for (a) the self-collected dataset and (b) the C-Town dataset.

performance is optimal when $K = 7$, with a CA of 91.84% and a silhouette coefficient of 0.752. The results show that as the network size and topology complexity increase, it is necessary to adjust the K value accordingly to achieve optimal performance.

3.2 | Comparison With SOTA Deep Graph Clustering Methods

To rigorously evaluate the performance of our proposed model, we conducted a comparative analysis against several SOTA deep graph clustering and unsupervised graph representation learning methods. These baselines represent the current leading

paradigms in learning from graph-structured data without labels. For methods that primarily learn node representations, we apply the K -Means algorithm to the learned embeddings to obtain the final clustering results.

The selected SOTA baseline models are:

3.2.1 | Graph Autoencoder (GAE)

A foundational framework for unsupervised graph representation learning. GAE learns node embeddings by training a GCN-based encoder to reconstruct the graph's adjacency matrix. It serves as a strong baseline for reconstructive feature learning.

TABLE 3 | Clustering performance under Enlarged City Scale and Fractal-like Topology scenarios.

Scenario	Number of clusters (K)	Clustering accuracy (CA) (%)	Silhouette coefficient
Baseline	5	94.77	0.791
Enlarged scale	4	88.45	0.712
	5	92.18	0.763
	6	93.05	0.778
	7	90.92	0.741
	5	89.67	0.728
Fractal-like topology	7	91.84	0.752
	8	90.11	0.735
	9	88.53	0.719

TABLE 4 | Performance comparison with SOTA baselines.

Model	Dataset	CA (%)	Silhouette coefficient	Key node coverage (betweenness) (%)	Key node coverage (flow betweenness) (%)	Key node coverage (current-flow closeness) (%)
GAE	Self-collected	89.15	0.675	88.52	85.23	86.15
	C-Town	88.31	0.642	86.90	83.72	84.53
DGI	Self-collected	91.82	0.721	90.11	88.31	89.03
	C-Town	90.59	0.693	89.42	86.52	87.87
GRACE	Self-collected	92.53	0.744	91.83	90.10	90.93
	C-Town	91.14	0.718	90.50	88.95	89.72
DCRN	Self-collected	93.16	0.762	92.52	91.47	92.11
	C-Town	92.07	0.725	91.71	90.24	90.86
SDC (proposed)	Self-collected	94.77	0.791	96.53	95.85	96.29
	C-Town	93.89	0.747	94.27	93.18	93.73

3.2.2 | Deep Graph Infomax (DGI)

A widely recognized method based on mutual information maximization. DGI learns node representations by maximizing the mutual information between local node embeddings and a global graph summary vector, ensuring the embeddings capture both local and global structural properties.

3.2.3 | Graph Contrastive Learning (GRACE)

A leading model representing the contrastive learning paradigm. GRACE generates two corrupted views of the input graph and learns node embeddings by maximizing the feature consistency of the same node across these two views. This approach has proven highly effective for learning robust representations.

3.2.4 | Deep Graph Clustering via Dual Correlation Reduction (DCRN)

A recent, powerful end-to-end deep graph clustering model. DCRN jointly learns embeddings and performs clustering while explicitly decorrelating features and cluster assignments to reduce redundancy and learn more discriminative representations.

These baselines were evaluated against our proposed method (SDC) on both the self-collected and C-Town datasets. Performance was measured using CA and the silhouette coefficient to assess both the correctness and the quality of the clusters.

The results presented in Table 4 clearly demonstrate the effectiveness of our proposed model. While all the SOTA baselines show strong performance, significantly outperforming the traditional methods listed in Section 3.1. As shown in Table 4, our proposed method consistently achieved the highest CA and silhouette coefficient scores on both datasets, demonstrating its superior performance over existing SOTA baselines.

3.3 | Optimization of Drainage Network Monitoring Points

The study first compares the number of monitoring points and the coverage of key nodes before and after optimization to verify the effectiveness of the proposed method in reducing monitoring point redundancy while simultaneously improving key node coverage. Figure 9 displays the findings. The number of monitoring points after optimization is reduced significantly. In the self-collected dataset, the average number of monitoring points decreases from 47 to 26. In the C-Town dataset, it decreases from

145 to 96. Meanwhile, the coverage of key nodes is significantly improved, from 73.42% and 67.84% to 96.54% and 91.89% in the two datasets, respectively. This indicates that the proposed method has a significant effect in reducing redundant monitoring points and improving the coverage of key nodes.

To verify the practical application effect of the proposed method, the study selects a mature residential drainage system as a practical test scenario, and analyzes the monitoring efficiency and flow error of the proposed method. It is also contrasted with other approaches. In Figure 10a, the average optimization times of KMC, silhouette coefficient, DEC, and AGNES are 44.72, 33.68, 28.12, and 22.36 s, respectively. The average optimization time of the proposed method is 16.85 s, which is shortened by 62.32%, 49.97%, 40.08%, and 24.62% compared with other methods, respectively. In Figure 10b, the average flow errors of the five methods are 4.76%, 4.08%, 3.25%, 2.27%, and 1.62%, respectively. The findings demonstrate that the proposed method significantly enhances drainage network monitoring performance due to its

increased monitoring efficiency and reduced flow inaccuracy in real-world applications.

To verify the real-time and reliability of the proposed method, the study analyzes the average response time and monitoring reliability of 10 monitoring points in a real environment. Table 5 displays the findings. After the optimization, the average response time of the monitoring points is shortened from 29.42 to 18.85 s, a decrease of 35.91%. The average monitoring reliability improved from 85.95% to 94.40%, an increase of 8.45 percentage points. The outcomes demonstrate the superiority of the proposed method in real-time monitoring by guaranteeing high dependability with a quicker reaction time.

3.4 | Robustness and Sensitivity Analysis

3.4.1 | Robustness Under Storm Scenarios

To evaluate the robustness of various methods under extreme conditions, the coverage of key nodes under normal and storm

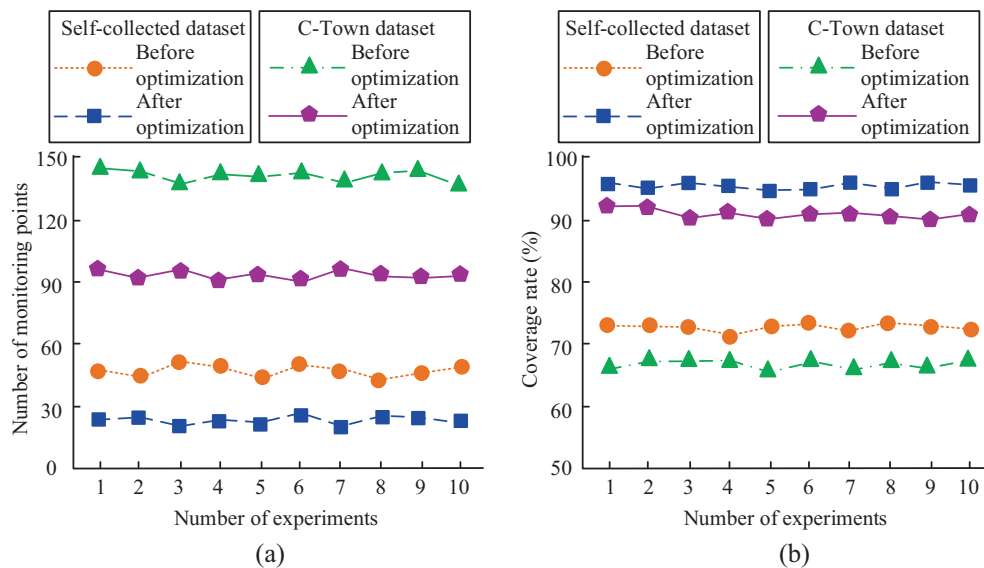


FIGURE 9 | Comparison of monitoring points before and after optimization including (a) number of points, (b) key node coverage rate.

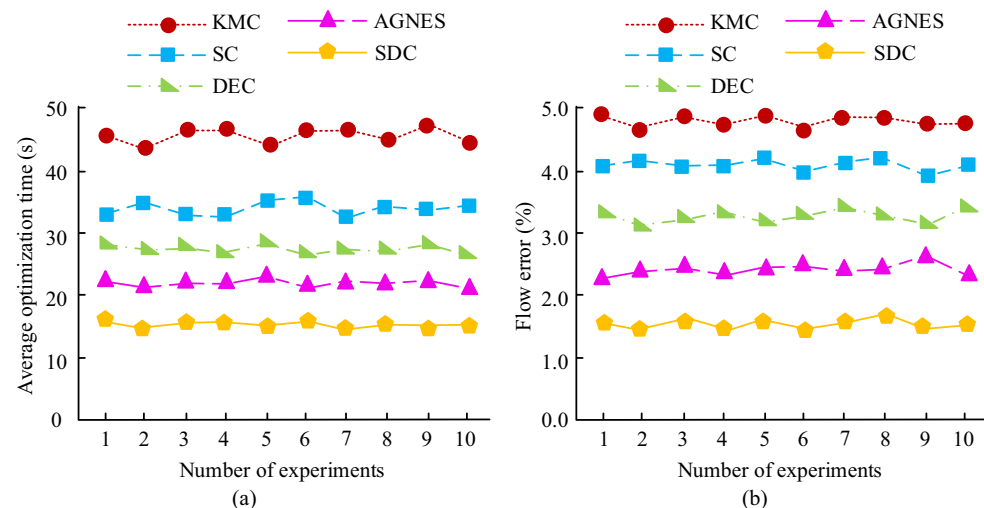


FIGURE 10 | Comparison of (a) average optimization time and (b) flow error of different methods.

conditions was analyzed in a self-collected dataset. The results are shown in Table 6. In Table 6, the coverage rate of KMC method decreased from 82.15% to 75.38%, a decrease of 8.24%. The SC method decreased from 85.60% to 79.21%; The DEC method decreased by 5.36%; The AGNES method decreased from 91.44% to 87.05%, a decrease of 4.80%; The SDC method proposed in this article performs the best, achieving a coverage rate of 96.54% under normal conditions and maintaining 93.21% under storm conditions, with only a decrease of 3.45%. The results showed that all methods exhibited a decrease in performance under storm conditions, but the SDC method not only maintained the highest coverage under all conditions, but also had minimal performance fluctuations, demonstrating optimal robustness and stability, and better adapting to the dynamic changes in hydraulic conditions in urban drainage networks.

3.4.2 | Sensitivity to Maximum Time Lag (M)

A sensitivity analysis was conducted by varying M from 5 to 25 time steps (equivalent to 25 to 125 min). For each value of M , the set of final selected monitoring points was recorded. The stability of the results was quantified by measuring the percentage of nodes that remained consistently selected across consecutive M values, indicating robustness. The results are summarized in Table 7. The results demonstrate that the selection stabilizes at $M = 15$, with no further changes observed for larger windows. This indicates that the relevant hydraulic delays in our

network clusters are captured within this window. Furthermore, the value $M = 20$ was ultimately chosen for all final experiments, as it was verified through hydraulic analysis to exceed the maximum expected travel time between any two nodes within a cluster, thereby effectively eliminating the risk of choosing a “pseudo-representative” node due to an insufficient lag window.

3.4.3 | Sensitivity to Model HYPERPARAMETERS

We analyzed the sensitivity of the proposed method to two key hyperparameters: the number of GCN layers and the dimensionality of the HL dimension. As shown in Figure 11a, the model performance, measured by CA, peaks at 3 GCN layers. Fewer layers appear insufficient to capture complex dependencies, while more than three layers can lead to over-smoothing, which degrades performance. Figure 11b shows that a hidden dimension of 64 provides the optimal balance between feature representation capacity and model complexity. These results provide empirical justification for our chosen parameter settings.

3.5 | Ablation Study

To validate the contribution of each key component in the proposed method, we conducted a comprehensive ablation study. We tested three variants of the proposed method: (1) w/o GCN, where the GCN module was removed and raw node features were

TABLE 5 | Average response time and monitoring reliability results for 10 monitoring points in a real-world environment.

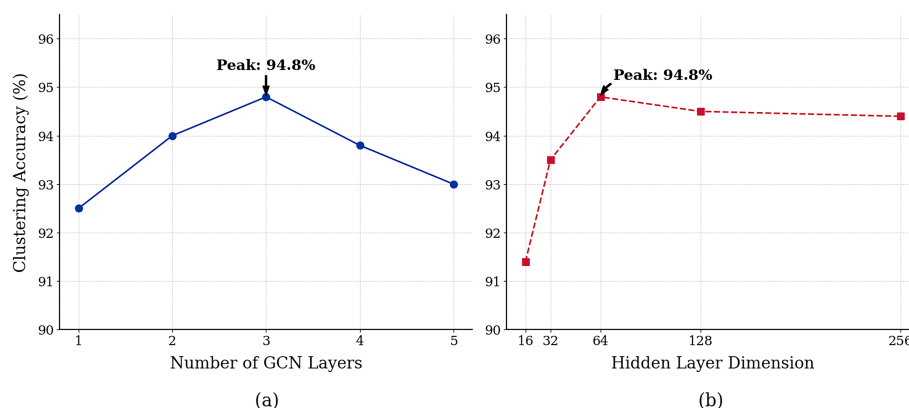
Monitoring point number	Original response time (s)	Optimized response time (s)	Original reliability (%)	Optimized reliability (%)
1	26.53	16.82	88.24	96.23
2	28.14	18.31	86.47	94.68
3	31.72	20.09	84.91	93.52
4	27.83	17.52	87.13	95.84
5	29.41	19.24	85.32	94.12
6	33.25	21.43	82.65	92.34
7	24.97	15.98	89.53	97.01
8	29.84	18.92	85.82	93.91
9	30.52	19.73	85.96	94.53
10	31.95	20.48	83.42	91.82
Average	29.42	18.85	85.95	94.40

TABLE 6 | Key node coverage under normal and storm conditions.

Methods	Normal conditions—key node coverage (%)	Storm conditions—key node coverage (%)	Coverage decline percentage (%)
KMC	82.15	75.38	8.24
SC	85.60	79.21	7.46
DEC	89.73	84.92	5.36
AGNES	91.44	87.05	4.80
SDC	96.54	93.21	3.45

TABLE 7 | Sensitivity analysis of monitoring point selection to the maximum time lag.

Maximum time lag (M)	Equivalent duration (min)	Number of monitoring points selected	Consistency with previous M (%)
5	25	28	N/A
10	50	26	85.7
15	75	26	100.0
20	100	26	100.0
25	125	26	100.0

**FIGURE 11** | Parameter sensitivity analysis for (a) number of GCN layers and (b) HL dimension on the self-collected dataset.**TABLE 8** | Ablation study results.

Model variant	CA (%)	Silhouette coefficient
Proposed (full model)	94.77	0.791
w/o GCN	87.15	0.624
w/o DNN	91.08	0.715
w/o Dual SS	89.42	0.688

used; (2) w/o DNN, where the non-linear transformation from the DNN module was omitted; and (3) w/o Dual SS, where the dual self-supervision mechanism was replaced with a standard KMC loss. The results on the self-collected dataset are presented in Table 8.

The results clearly indicate that removing any component leads to a significant degradation in performance. The largest drop occurs with the removal of the GCN module, underscoring the critical role of topological feature extraction for accurately representing the network structure. The dual self-supervision mechanism also proves essential for achieving high-quality, well-separated clusters, as its removal results in a substantial decrease in both accuracy and silhouette score.

4 | Discussion

The superior performance of our unsupervised method can be attributed to its graph-native architecture, which directly processes the network's inherent topology. Unlike traditional

baselines such as DBN, CNN, and RNN, which require sequentializing node embeddings and thereby lose critical structural context, our GCN-based approach learns from the most relevant structural information from the outset. This direct processing of the non-Euclidean pipe network topology enables a more comprehensive feature representation, leading to both higher accuracy and faster convergence. Furthermore, our ablation study (Table 8) confirmed that the innovative dual self-supervised clustering module was highly effective, learning discriminative features without external labels to produce more compact and well-separated clusters.

The practical implications of these performance gains are substantial. A 36% reduction in average response time, for instance, translates to faster detection of critical events like pipe blockages or overflows. This enables municipal operators to dispatch maintenance crews more quickly, potentially preventing sanitary sewer overflows that pose significant public health risks and lead to costly environmental fines and regulatory penalties. Similarly, reducing the number of sensors by over 40% while improving key node coverage leads to significant capital and operational savings. To illustrate, for a district with 500 potential monitoring points, a 40% reduction eliminates the need to purchase, install, and maintain 200 sensors. This could translate into capital and operational savings exceeding hundreds of thousands of dollars, making advanced network monitoring financially viable.

This work offers a methodological advancement by presenting an unsupervised framework that learns directly from network topology, thereby circumventing the need for historical failure data required by many supervised models. Previous studies, such

as the work of Xu et al. [7] and Liu et al. [8], also applied GCNs, but their supervised learning paradigm requires extensive historical failure or pressure data, which are often unavailable or incomplete in many real-world systems. By contrast, our unsupervised framework circumvents this critical limitation by learning directly from the network's inherent topology, making it far more scalable and readily deployable in data-scarce environments. In addition, compared with studies focusing only on pattern recognition or a single prediction task, this study directly addressed the more fundamental engineering decision-making problem of "where to monitor." By combining advanced graph representation learning with unsupervised clustering, this study provides an end-to-end solution for optimal resource allocation, which holds greater practical significance.

The findings of this study translate into significant economic and operational advantages for urban water management. The optimized monitoring point layout not only achieved higher coverage of critical nodes (more than 20% improvement) with fewer sensors (44.7% reduction in self-collected dataset), but also dramatically improved the operational efficiency of the monitoring system. The average response time was reduced by almost 36% (from 29.42 to 18.85 s, as detailed in Table 5). This meant that the system could issue early warnings for abnormal events, such as overflows and blockages in the pipeline network, much earlier. This provides valuable time for emergency response and dispatch, reducing the risk of urban flooding and environmental pollution.

However, this study also has some limitations. For one, the generalization ability of the model needs to be further verified. The self-collected dataset of this study is mainly derived from residential neighborhoods, whose pipe network topology is relatively regular. For more complex pipe network systems, such as industrial areas, commercial areas, or cities with high topographic relief, the model's adaptability needs to be tested by a wider range of datasets. A promising future research direction is to develop a foundational model for urban drainage networks, pre-trained on a diverse corpus of network archetypes. Such a model could offer powerful zero-shot generalization to new, unseen systems, significantly reducing deployment costs. The normalized adjacency matrix is theoretically topology-agnostic, as it operates on relative node connectivity rather than absolute spatial configurations. This fundamental property enables consistent feature extraction across diverse network layouts, including industrial and mountainous topologies, without requiring retraining. Second, the current model primarily considers static pipe network topology. It does not sufficiently account for the real-time impact of dynamic hydraulic conditions, such as flow fluctuations caused by changes in rainfall intensity, on node importance. Future research should focus on developing dynamic ST-GCNs. A key challenge will be to effectively fuse sparse, real-time hydraulic data with the static network topology, enabling the model to shift from static optimization to a dynamic, adaptive optimization strategy that responds to events like rainfall in real-time.

5 | Conclusion

This study introduced and validated a novel unsupervised framework for optimizing drainage network monitoring points,

leveraging the strengths of GCNs and SDC. Our method demonstrated superior performance, consistently outperforming SOTA alternatives. These technical achievements translate directly into profound operational benefits: the number of required monitoring points was reduced by over 40%, while critical node coverage exceeded 90%, leading to a nearly 36% faster response time to critical events. In conclusion, this study offers more than an optimized algorithm; it provides a robust blueprint for advancing a paradigm shift in urban water management. By enabling the move from static, experience-based monitoring to an intelligent, data-driven framework, our work empowers municipalities to build the resilient and sustainable water infrastructure of the future.

Author Contributions

Yaole Qi: conceptualization, methodology, software, writing – original draft. **Qiang Li:** supervision, project administration, funding acquisition, writing – review and editing. **Guangrong Cao:** data curation, formal analysis, validation. **Gang Li:** investigation, resources, visualization. **Lei Zhang:** software, methodology, investigation. **Bo Yang:** data curation, formal analysis, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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